

Chapter 1

Functions

This week begins with one of the central habits of precalculus: looking at a set of points and asking whether there is a rule behind them. Sometimes the points are just a collection of measurements. Sometimes they line up in a pattern clearly enough that a formula can describe them. The vocabulary of *relations* and *functions* gives us a way to separate those two situations.

A relation is a collection of points. A function is a relation in which each input determines a single output. That condition matters because it is the condition that makes prediction possible: once the input is known, the output is determined. Much of this course is about finding predictable rules in data, recognizing familiar shapes, and then reshaping those familiar functions into models of the form

$$A f(b(x - h)) + k.$$

The algebra is important, but the guiding question is geometric: what shape are we looking at, and how has it been moved?

1.1 Relations as Point Clouds

Before we start with formulas, it helps to think about data. Record, for every student in a class, their height and their shoe size. Each student becomes a pair of numbers: an input (height) and an output (shoe size). Plotted, these pairs form a cloud of dots.

At this stage there is no formula yet. The cloud is simply the information we have. It may suggest a pattern, or it may not. A relation is the mathematical object that lets us talk about the cloud before deciding whether it deserves to be called a function.

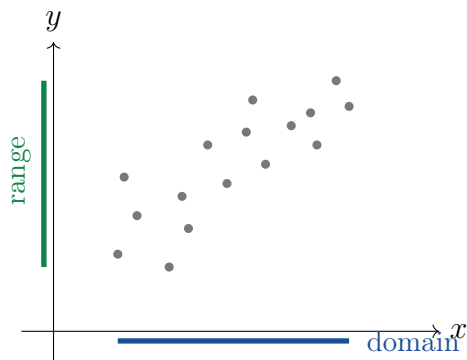
Definition: Relation. A *relation* is any set of ordered pairs (x, y) . The set of all first coordinates is the *domain*; the set of all second coordinates is the *range*.

A relation can be specified by a list, a table, an equation, or a graph. These are four ways of naming the same set of pairs, and moving among those forms is one of the basic skills of the course. The list

$$\{(1, 2), (3, 5), (3, -1), (4, 0)\}$$

is a relation with domain $\{1, 3, 4\}$ and range $\{2, 5, -1, 0\}$. So is the set of all (x, y) satisfying $x^2 + y^2 = 1$; that relation is a circle, an infinite set of points packed densely enough to read as a curve.

The picture. A relation is a cloud of dots suspended above the x -axis. The domain is the shadow the cloud casts straight down onto the x -axis; the range is the shadow it casts sideways onto the y -axis. Domain and range are the two shadows of the cloud.



Example 1.1.1. (*reading domain and range from a list*) For $\{(-2, 4), (0, 0), (1, 1), (2, 4)\}$ the domain is $\{-2, 0, 1, 2\}$ and the range is $\{0, 1, 4\}$. The value 4 appears twice in the range but is listed once; a range is a set, so repeats collapse.

Notice that the example did not require a formula. A finite list of points is already a relation. The formula enters only when we try to describe many points at once, or when we want to predict values that were not listed.

1.2 When a Relation Is a Function

Now we add the extra condition that makes a relation predictable. A general relation may pair the same input with several outputs. That is allowed for a relation, but it is not allowed for a function.

A function is stricter. Once an input has been chosen, the function gives one output and only one output. This is the reason functions are useful as models: they let us ask, “What happens when the input is this?” and expect a definite answer.

Definition: Function. A *function* is a relation in which each input is paired with exactly one output. Equivalently, no two distinct pairs share the same first coordinate.

The relation $\{(1, 2), (3, 5), (3, -1), (4, 0)\}$ is not a function: the input 3 is paired with both 5 and -1 . Deleting either pair makes it a function. The circle $x^2 + y^2 = 1$ is not a function either; the input $x = 0$ yields both $y = 1$ and $y = -1$. The issue is not whether the relation is interesting or geometric. The issue is whether each input selects a single output.

Key idea. A function assigns to each input exactly one output. The domain is the set of inputs for which the assignment is defined; for each such input the output is determined.

When a function is given by a rule, $f(x)$ denotes the output at input x . The symbol $f(2)$ means the output of f at input 2, a single number. It does not mean f multiplied by 2. This notation is compact, but it is also literal: whatever is inside the parentheses is the input being fed into the rule. **Example 1.2.1.** (*evaluating a function*) Let $f(x) = x^2 - 3x + 1$. Then

$$f(2) = 4 - 6 + 1 = -1, \quad f(0) = 1, \quad f(-1) = 1 + 3 + 1 = 5.$$

And $f(a+1) = (a+1)^2 - 3(a+1) + 1 = a^2 - a - 1$ after expanding. Evaluating at an expression means substituting the entire expression for x .

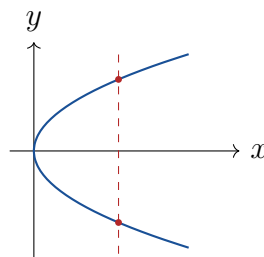
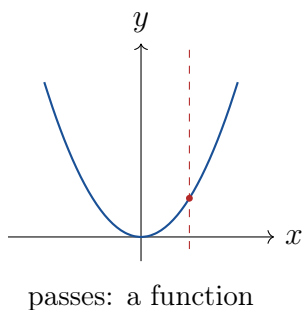
The last part is the one to practice slowly. The input is not just the symbol a ; the input is the whole expression $a+1$. Parentheses protect that input while the function rule acts on it.

Common pitfall. $f(a+b)$ is almost never $f(a) + f(b)$. For $f(x) = x^2$, $f(1+2) = f(3) = 9$, while $f(1) + f(2) = 1 + 4 = 5$. The function acts on the entire input at once, not on its parts separately.

1.2.1 The Vertical Line Test

The function property also has a graphical reading. A graph is made of points, and points with the same x -coordinate lie vertically above one another. So the statement “each input gives one output” means that above any point on the x -axis the graph has at most one point. A vertical line is exactly the set of points sharing one x -value, which gives the test directly.

Key idea. *Vertical line test.* A graph represents a function if and only if no vertical line crosses it more than once.



fails: not a function

Example 1.2.2. (*applying the test*) The graph of $y = |x|$ is a V opening upward; every vertical line hits it once, so it is a function. The graph of $x = y^2$ is a sideways parabola; the vertical line $x = 1$ hits it at $(1, 1)$ and $(1, -1)$, so it is not a function.

1.3 Fitting a Function to a Cloud

The two ideas now meet. Real data usually forms a point cloud with noise; it is almost never an exact function. Two people with the same height may have different shoe sizes, and repeated measurements of the same process may not agree perfectly. That does not make functions useless. It means we use them carefully.

A function can be laid through the cloud as a *model*: a predictable rule that follows the trend closely enough to summarize the data and make reasonable forecasts. The model is not a claim that the data points magically lie on a perfect curve. It is a controlled simplification, chosen because it captures the main pattern.

Key idea. A point cloud is the data. A function fit through it is a *model*. The model is not the data; it is the predictable rule chosen to represent the data so that one can compute and forecast with it.

The first decision is the *shape*. A cloud rising in a straight band calls for a linear model. A cloud curving upward like a bowl suggests a parabola. A cloud that climbs quickly and then levels off suggests a square-root or logarithmic shape. This is why we build a catalog of *parent functions*. They are the basic shapes we learn to recognize before the algebra becomes complicated.

Once the shape has been chosen, the remaining work is to move and resize it. That is where transformations enter.

Interactive. A hidden transformed parent function generates a noisy point cloud. Identify the parent-function family, then tune $y = A f(b(x - h)) + k$ until the curve threads the cloud. The fit score reports the root-mean-squared error between curve and data; the hidden answer is deliberately approximate because the cloud carries noise.

https://bobovski66.github.io/Precalculus/demos/function_fit.html

Example 1.3.1. (*choosing a shape*) A cloud trends upward and bends, steep at first and then flatter. A straight line undershoots the early rise and overshoots the later leveling. The square-root shape $f(x) = \sqrt{x}$ has exactly this profile, fast and then flattening, so $\sqrt{x}$ is the parent to start from; the fitting work is sliding and stretching it onto the cloud.

1.4 Transformations: Bending One Curve into Another

So far, a function has been a rule that assigns one output to each input. Now we use functions as shapes. The point is not only to evaluate a formula, but to understand how changes in the formula move the graph.

Every function in the library can be reshaped by four operations, all captured in a single expression. Given a parent function f , form

$$g(x) = A f(b(x - h)) + k.$$

Each constant controls one geometric action. When we fit a transformed parent function to data, we are usually not inventing a new formula from nothing. We choose the parent shape first, then use A , b , h , and k to move and resize it until it matches the cloud reasonably well. The constants become geometric controls.

Key idea. The four constants of $g(x) = A f(b(x - h)) + k$.

- k : vertical shift. Adds to the output, moving the graph up ($k > 0$) or down ($k < 0$).
- h : horizontal shift. Moves the graph right ($h > 0$) or left ($h < 0$).
- A : vertical stretch by factor $|A|$; if $A < 0$, also a flip across the x -axis.
- b : horizontal stretch by factor $1/|b|$; if $b < 0$, also a flip across the y -axis.

1.4.1 Outside versus inside

The reliable bookkeeping is the outside/inside rule. Constants applied *outside* f (A and k) act on the output, so they move the graph *vertically*. These usually behave the way the notation suggests: $+k$ raises the graph, and A scales its height.

Constants applied *inside* f (b and h) act on the input before the parent function sees it. Because they change the input, they move the graph *horizontally*, and the direction can feel backwards at first: $x - h$ shifts *right* by h , and multiplying the input by b compresses the graph horizontally by a factor of $1/|b|$.

The picture. Outside changes act on the picture after it is drawn: raise it, scale its height. Inside changes act on the input before f receives it, so they warp the horizontal axis, and warping the input runs opposite to intuition. The parent produces its value at input 0; the transformed graph reaches that value where $b(x - h) = 0$, that is, at $x = h$. This is why $x - h$ moves the graph right rather than left.

Common pitfall. The horizontal shift is $x - h$, so $g(x) = f(x - 3)$ moves the graph *right* by 3, not left. The minus sign is built into the form. Read $f(x + 3)$ as $f(x - (-3))$, so $h = -3$ and the shift is left by 3.

Example 1.4.1. (*a single vertical shift*) With parent $f(x) = x^2$, the function $g(x) = x^2 + 3$ is the parabola lifted up 3 units. The vertex moves from $(0, 0)$ to $(0, 3)$; every output gains 3.

Example 1.4.2. (*a single horizontal shift*) With $f(x) = x^2$, the function $g(x) = (x - 2)^2$ slides the parabola right 2 units. The vertex moves from $(0, 0)$ to $(2, 0)$. Check: $g(2) = 0$, the old value at the old vertex input 0.

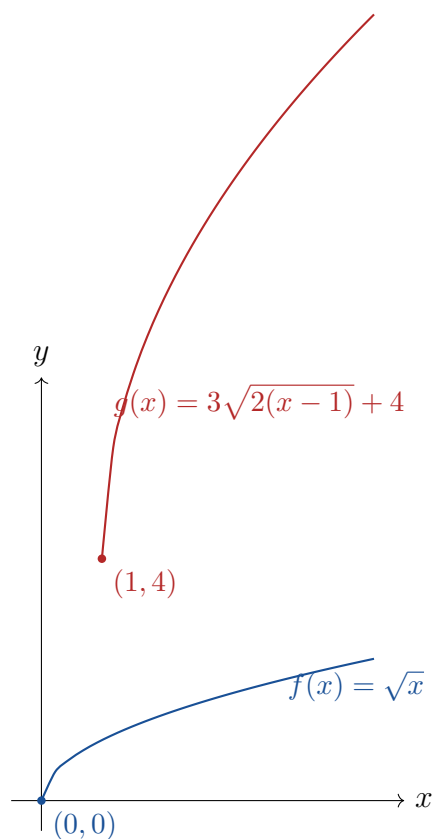
Example 1.4.3. (*stretch and reflect together*) With $f(x) = x^2$, the function $g(x) = -2x^2$ has $A = -2$: stretch vertically by 2 (twice as tall) and flip across the x -axis (opens downward). The point $(1, 1)$ on the parent becomes $(1, -2)$.

Example 1.4.4. (*all four at once*) Let $f(x) = \sqrt{x}$ and

$$g(x) = 3\sqrt{2(x - 1)} + 4.$$

Read off $A = 3$, $b = 2$, $h = 1$, $k = 4$. Starting from $\sqrt{x}$: compress horizontally by $\frac{1}{2}$, stretch vertically by 3, shift right 1, shift up 4. The parent begins at $(0, 0)$; the transformed curve begins at $(h, k) = (1, 4)$, since the inside is zero at $x = 1$ and the $+k$ raises the value to 4.

The useful habit is to look for a landmark point on the parent graph, such as a vertex, corner, endpoint, or intercept. Transform that point first. Once a few landmarks are in place, the rest of the graph is much easier to sketch.



1.4.2 Order of operations on the graph

When several transformations combine, the safest method is to follow one point on the graph. The order mirrors the evaluation of $g(x)$ at a number: work from the input outward. Apply the inside first (horizontal: the b compression, then the h shift), then the outside (vertical: the A stretch or reflection, then the k shift). Following one landmark point of the parent through each step keeps the order honest and prevents the horizontal changes from being applied backwards. **Example 1.4.5.** (*tracking a point*) Take $f(x) = \sqrt{x}$ and the landmark $(4, 2)$ on it. Apply $g(x) = 3\sqrt{2(x-1)} + 4$ to that point's coordinates:

$$x : 4 \xrightarrow{\div b} 2 \xrightarrow{+h} 3, \quad y : 2 \xrightarrow{\times A} 6 \xrightarrow{+k} 10.$$

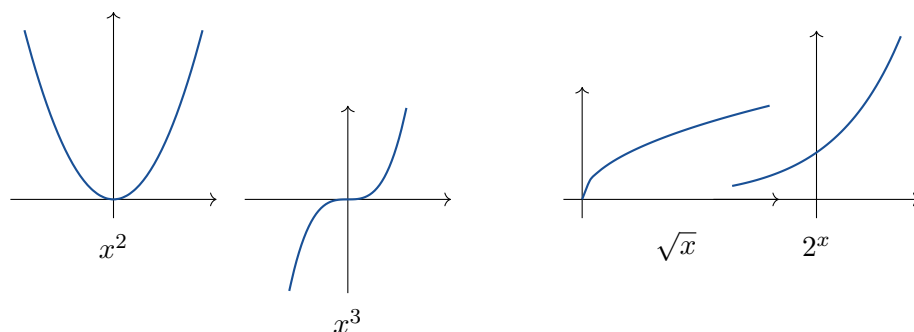
So $(4, 2)$ lands at $(3, 10)$. Confirm directly: $g(3) = 3\sqrt{2 \cdot 2} + 4 = 3 \cdot 2 + 4 = 10$. The two routes agree.

Common pitfall. The input coordinate transforms by the *inverse* operations in *reverse* order: to locate where an x lands, undo the inside. Dividing by b before adding h matches the algebra of $b(x - h)$; reversing those two steps misplaces the point.

1.5 The Library of Parent Functions

Fitting is limited by the shapes one knows. The following parent functions recur throughout the course. The goal is not to memorize them as isolated formulas, but to recognize their visual behavior: straight, bowl-shaped, V-shaped, S-shaped, flattening, branching, or rapidly growing. Once the rough shape is recognized, the transformation parameters can do the more precise fitting.

Name	Rule	Shape in a sentence
Constant	$f(x) = c$	flat horizontal line
Linear	$f(x) = x$	straight line through the origin, slope 1
Absolute value	$f(x) = x $	V opening up, corner at origin
Quadratic	$f(x) = x^2$	bowl opening up, vertex at origin
Cubic	$f(x) = x^3$	S-curve, flat at origin
Square root	$f(x) = \sqrt{x}$	half-parabola, fast then flattening
Reciprocal	$f(x) = 1/x$	two branches, axes as asymptotes
Exponential	$f(x) = 2^x$	flat then explosive growth



Example 1.5.1. (*matching a cloud to a parent*) A cloud lies entirely in the first quadrant, rising steeply near zero and then bending to a gentle climb. Among the parents, only $\sqrt{x}$ has this steep-then- gentle profile beginning at the origin. Start from $\sqrt{x}$ and fit A, b, h, k .

At this stage, a good first guess is more valuable than a perfect answer. Choose the parent family from the overall shape, then adjust the constants to improve the fit.

1.6 Exercises

Exercises 1.6

- 1.6.1. For the relation $\{(-1, 3), (0, 0), (2, 3), (2, -3), (5, 1)\}$, state the domain and range, and decide whether it is a function. Justify.
- 1.6.2. Decide whether each graph could pass the vertical line test, and say what the test is checking: (a) a circle; (b) a downward parabola; (c) the letter “S” lying on its back; (d) a single steep upward line.
- 1.6.3. Let $f(x) = 2x^2 - x + 3$. Compute $f(0)$, $f(-2)$, and $f(t+1)$, simplifying the last completely.
- 1.6.4. Give a relation that is *not* a function, then change exactly one ordered pair to make it a function. Explain why your change works.
- 1.6.5. Starting from $f(x) = x^2$, write the rule for the function obtained by shifting left 4, stretching vertically by 3, and shifting down 1. Then state the vertex.
- 1.6.6. Describe in words every transformation taking $f(x) = |x|$ to $g(x) = -\frac{1}{2}|x + 2| + 5$, in the correct order, and give the coordinates of the corner of g .
- 1.6.7. The point $(9, 3)$ lies on $f(x) = \sqrt{x}$. Find where it lands under $g(x) = 2\sqrt{x-1} + 4$ by tracking the coordinate through each step, then confirm by direct substitution.
- 1.6.8. Explain, using the outside/inside rule, why $f(x-3)$ shifts the graph right while $f(x) + 3$ shifts it up. Why do the two “+3” moves go in such different directions?
- 1.6.9. A point cloud rises in a nearly straight band but with clear scatter. Explain the difference between the cloud and a linear function fit through it, and state what the word *model* means in this context.
- 1.6.10. For $g(x) = A f(b(x-h)) + k$, suppose $A < 0$ and $b < 0$. Describe both reflections that occur and name the two axes involved.
- 1.6.11. A cloud opens upward like a bowl with its lowest point near $(2, -3)$. Propose a parent function and initial values of A, b, h, k to start a fit, and explain each choice.
- 1.6.12. Open the Function Fit Lab. Generate a cloud, identify the parent family, and record the four constants you settled on with the resulting fit score. State in one sentence which constant was hardest to determine and why.